

VisionQC

Adaptive AI Quality Control HUD

1. Project Overview

VisionQC is an adaptive, vision-based quality inspection system built to eliminate the biggest bottleneck in AI-driven quality control: retraining. Instead of relying on a fixed model trained for one specific defect, VisionQC uses an open-vocabulary detection engine that responds directly to natural-language prompts. An operator can type a new inspection target — “scratch,” “bruised spot,” “hole,” or “mold” — directly from an immersive heads-up display, and the system begins detecting that exact defect on the production line within seconds.

The platform closes the loop end-to-end: a camera captures the product, an automation layer coordinates AI inference and decision-making, embedded hardware physically removes defective items, and every detection is logged for traceability. The result is a flexible, low-code inspection system that can be redeployed across different products, defect types, and production lines without writing a single line of new AI code — making advanced computer vision accessible to small and mid-sized manufacturing lines.

2. Pain Points & Solutions

Pain Point	VisionQC Solution
Retraining an AI model for every new defect type is slow and costly	Open-vocabulary detection adapts instantly via a text prompt — no retraining
Manual inspection is inconsistent and prone to fatigue-related errors	Automated, always-on AI detection delivers consistent, objective results
Rigid systems can't adapt to different products on the same line	One flexible pipeline handles multiple products and criteria on demand

3. Tech Stack & Key Features

- **Prompt-based defect detection** — instantly change inspection targets via text, with zero AI retraining
- **Real-time visual overlay** — live video feed with AI-detected bounding boxes rendered on the Pico 4 HUD
- **Automated rejection** — defective items are automatically flagged and physically removed via ESP32-driven actuators
- **Confidence-based filtering** — a configurable threshold ensures only high-confidence detections trigger an action
- **Live data logging** — every detection is recorded with timestamp, defect type, and confidence score
- **No-code, modular workflow** — n8n orchestrates the full pipeline, making the system easy to extend or reconfigure

4. System Flow Architecture

Stage	Tech Stack	Function
Capture	Raspberry Pi + Camera	Streams live video and captures frames on trigger or timed interval
Orchestrate	n8n	Routes images and prompts, evaluates confidence scores, triggers downstream actions
Detect	NVIDIA LocateAnything	Open-vocabulary AI detects defects from the prompt and returns bounding box coordinates
Act	ESP32	Triggers reject actuators (servo/relay) and reads sensor input from the production line
Visualize	Pico 4 Web HUD	Displays the live feed with AI-detected bounding boxes; lets operators update prompts instantly
Log	Google Sheets / MySQL	Records timestamp, defect type, and confidence score for every detection